Why Forward Time, Complexity and Entropy? Reversing Infinite Time, Simple Laws, and Evolving Minds

Gary Abraham Bernstein

Independent Researcher · ORCID: 0009-0009-1761-2867

Abstract

Why do we experience forward time? Information can propagate from a supplied initial state through a history with no last event. Complete backward propagation, if each step must receive an already-produced input from the next-later step, has no final source from which to begin. The paper argues that this matters if experience requires such initiated production. Even where reversal is possible, its probability weight may be lower. Reversing a real billiard break requires scattered light, sound and heat to return in coordination. Under a fixed binary description measure, each unavoidable extra bit halves the reverse description's weight; whether that coordination requires extra bits must be established. Observer selection adds a developmental requirement: a beginning without the organization needed for later minds must develop it. Biological development uses energy gradients while total entropy increases. Hidden coordination may also restrict reliable learning, although a fair comparison must allow the observer's records and orientation to reverse too. Beginning with a hypothetical coin flip between directions, these arguments form a cumulative case for forward experience. The paper separates the exact conditional results from the further comparisons needed to establish the proposed probability preference.

1. The proposed explanation

We experience forward time because information can propagate forward from an initial state, whereas a complete backward propagation has no final state from which to begin. Each backward step would need information handed to it by the next-later step. Without an originating input, none of those steps produces its output. This is the central argument proposed here.

The issue is the production of the information constituting experience, not somebody reading a completed list in reverse. Reversing the notation leaves that production question unanswered. The argument therefore asks whether a history capable of experience must have a rooted chain of information-producing dependencies.

Observer development provides the second part. A simple beginning that lacks the required organization must generate it if it is to contain observers like us. Those observers also need usable relations between what they can sense, what they can learn and what happens next. A world in which important events depend on hidden, precisely coordinated arrivals may constrain the conditions

under which learning and adaptation develop. Reversed splashes make that concern concrete. In our physical history, biological development uses available energy and increases total entropy.

2. There is no last event from which to start

Consider events indexed 0, 1, 2, 3, and so on. Forward generation can supply event 0 and then obtain event n+1 from event n. The sequence can continue indefinitely. It needs no completed infinity before a finite event occurs.

Now attempt to generate the same complete sequence in the opposite dependency direction. Event n requires information from event n+1; that event requires information from n+2; and so on. If the rule can operate only after its input has been produced, and no input is supplied independently, no operation is initially enabled. Applying more copies of the rule supplies no first input. At every finite stage, no event has been produced.

A final event could seed such a reverse construction in a finite history. The endless history has no final event. Under the stated generation rules, its complete reverse construction therefore cannot begin. This is an exact result about initiated propagation.

The result applies to a sequence with a first event and no last event. An integer-indexed history has neither endpoint, while a finite history has both. Establishing that the rooted endless form is necessary for our physical history remains part of the proposed account.

3. The premise connecting sequence order to experience

The argument requires that information constituting an experience have a rooted production history: trace its generating dependencies back far enough, and they reach supplied information rather than continuing without an origin. It further requires that an opposite arrow for the same complete history obtain its information through the reverse dependencies, without an independently supplied state halfway through.

With those requirements, the conclusion follows. Every backward step waits for input, and no final event supplies it. A local backward calculation supplied with a state such as event 3 uses an extra input; it does not show how the full backward history originates. This answers the objection that writing 3, 2, 1 alone initializes the reversed universe.

There is nevertheless another objection. A mathematical structure can satisfy relations without being generated by propagating information through them. For example, x_n=n satisfies both x_(n+1)=x_n+1 and x_n=x_(n+1)−1 for every natural-number index n. The backward relations need no last value at infinity to be jointly specified. The formula describes the whole arrangement; it is not a backward construction of it.

The question is therefore whether such jointly existing relations can constitute experienced succession, or whether experience requires the rooted production demanded above. Identifying reality with mathematical structure does not alone decide between those accounts. A block need not be assembled from an endpoint merely to exist as a structure. If experience requires more than joint specification, that additional requirement must be explained within the structural account.

This is the decisive task for the proposed arrow. Show why a record of an experience cannot be instantiated by the relevant local organization alone, and why its dependence must have the specified origin and orientation. Then the missing final source rules out the opposite construction. The endpoint argument supplies that last step; the account of experience must supply the bridge.

Why We Remember the Past and not the Future: Records and the Arrow of Consciousness

A memory contains information about an interaction. The relevant question is whether forming that record requires the privileged direction of production proposed above. It cannot be answered merely by naming one state input and another output.

A simple example isolates the issue. Let x be a source bit and r a memory bit. The transformation (x,r) $\rightarrow$ (x,r XOR x), where XOR is addition modulo two, is its own inverse. Starting with r=0 produces a record r =x while retaining x. Applying the transformation again restores the blank memory. Recording information therefore need not be a one-way mathematical operation (Bennett, 1973).

The initial memory matters. A blank memory produces a faithful copy; an independent, uniformly distributed memory bit produces no statistical information about x in its final value. These probabilities may describe entirely deterministic trials. The same reversible rule can therefore make a useful record or an uninformative one depending on what information was supplied to it.

This supports a specific part of the present argument: a rule alone does not supply its starting information. It does not establish that conscious processing can occur only in one temporal orientation. A two-bit recorder is not a proposed conscious system. To reach that stronger conclusion, an account of experience must show why its required information production has the rooted direction described above. A proof of that claim would strengthen the main arrow argument directly.

4. Why observer complexity must develop

Simple laws can remain unchanged while the structures they generate acquire the organization required for observers. Suppose the starting state lacks that organization and a later state contains observers who require it. The history

must then develop the missing organization. This is a necessary consequence of those premises, not merely a possible outcome. The growing quantity is the observer relevant organization of the generated states; the law and the shortest description of the history can remain simple.

Observer conditioning supplies the second part of the argument. Histories that never contain the observers whose evidence is being explained have zero weight for that evidence. Within the class that begins below the required organization and later produces those observers, development is necessary. Comparing that class with histories that already contain observers at the chosen boundary is a separate selection calculation. Simple laws, simple total description and an initially unorganized state are three different premises.

There must be a first stage at which the required organization is present, if stages are indexed by the natural numbers. This establishes a crossing of the observer's requirements. It establishes neither growth at every step nor indefinite growth afterward. Organization may rise and fall many times; a mature observer can operate within a bounded range.

Two questions remain. Does economical total specification favor a starting state below the required organization, rather than one with a short description that already contains observers? And is the observer criterion independently justified? Defining an observer as something that developed from a simple beginning would make the conclusion circular.

Reversed splashes and the conditions for minds to evolve

A billiard break scatters balls, sends sound through the room and transfers energy into the table and surroundings. Light scattered from the event travels outward. A complete reversal would require the relevant light, sound and microscopic motions to return in coordination as the balls reassemble. Reversing only the balls would leave the rest of the event unreversed. To a learner who cannot observe that incoming coordination, the assembly can look like a sudden coincidence. The proposed obstacle is missing advance information, not merely slow thinking.

That appearance need not involve chance or any failure of the laws. It can arise because the learner cannot access the detailed correlations on which the event depends. Nor does a single ideal billiard collision establish an asymmetry: complete positions and momenta determine both directions. The issue is whether limited observations support useful predictions across the environments in which minds must develop.

Evolution requires organisms to survive and reproduce; learning is one way of doing so. For minds that use learned predictions, a useful response must keep working across some range of encounters. If tiny, unseen differences repeatedly determine whether scattered motions assemble into a threat, the same visible cue may support no reliable response. The proposed disadvantage is therefore

a restriction on the conditions supporting the evolution of effective learners. More processing speed alone cannot recover information their observations never supplied. Reversed histories are not all unpredictable; the argument concerns histories whose consequential coordination remains inaccessible to the learner.

Prediction and understanding afterward

The information limit can be stated without a model of the whole universe. Suppose two possible situations supply exactly the same observations and memory to a learner, but successful action requires different, incompatible responses. A decision rule using only that information must give the same response, or the same distribution of responses, in both situations. It therefore cannot guarantee success in both. Additional computing power does not remove this limit; additional distinguishing information can.

This is where understanding afterward differs from prediction. Later evidence may distinguish situations that looked identical when action was required. The learner can then reconstruct what happened, but that reconstruction was unavailable to the earlier decision. If important encounters repeatedly have this form, the learner repeatedly has to infer the relevant cause after its consequence has arrived. An outcome produced by many possible inputs does not, by itself, identify which input occurred. Reconstruction needs further evidence and may remain ambiguous.

The example 1+1=2 makes the distinction compactly: the sum alone does not determine the summands. The alternatives are underdetermined, not thereby random or uncaused. Nor is the example a proof that microscopic time evolution loses information. A complete reversible state can retain distinctions absent from the learner's observations. The proposed evolutionary obstacle is repeated dependence on information unavailable before a consequential choice. To establish a directional preference, that obstacle must persist more often or more severely in the reverse class under the comparison's stated conditions.

There is relevant prior work. Mlodinow and Brun (2014) relate the direction of memory to the thermodynamic arrow by requiring records to remain useful when the recorded system varies, without specially matching the memory to each case. Their argument allows reversible, nondissipative memories. Heinrich, Knopp and Päs (2015) compare agent survival in a toy environment of scattering or converging rocks. Their model illustrates a possible evolutionary disadvantage; its restricted agents and environment do not establish a general impossibility. The present proposal connects these questions to the conditions under which observer development occurs.

The comparison must include the observer. Reversing an environment while leaving a learner's recording processes unchanged differs from reversing a complete observer–environment history. In a time-reversal-invariant model, the latter transforms the records and their formation too. One cannot assume that the transformed observer continually encounters surprises merely because its movie

looks strange to us. The useful claim concerns alignment between an observer's records, adaptation and environment; reversing a coordinate label cannot defeat it.

To establish fewer viable developmental histories, specify which starting states are allowed, how they are weighted, and what sensing, memory and survival require. Compare the weight of histories that develop reliable learners in each orientation. Allow reverse-oriented learners to adapt, rather than testing only a forward-trained learner on reversed data. A disadvantage that remains would strengthen the explanation even if reverse experience is possible. Exact reversal pairs trajectories in reversible models; it does not itself make one class smaller. Any difference must come from the stated conditions and observer requirements. Narrower successful conditions also need not mean longer descriptions: a compact rule can specify extensive coordination.

5. Why simple laws enter the explanation

A short rule can generate an elaborate state without listing every constituent separately. Here complexity means the organization needed for the observers being explained. It need not be the number of incompressible bits in a shortest description, and its growth does not require the laws themselves to change.

Following the program-weighting approach of Solomonoff (1964), the simplicity proposal weights a complete description of length L by 2^(−L), using one stated prefix-free binary language. Complete programs in such a language are distinguishable without one being a prefix of another. Under this weighting, a description k bits shorter receives 2^k times the individual weight. This is the precise sense in which shorter descriptions are exponentially favored. It is a consequence of the chosen weighting, not a count showing that there are more short programs than long ones.

Count the starting state as well as the laws. A short law coupled to an unexplained elaborate beginning may be a long total explanation. Conversely, an elaborate-looking beginning might itself have a short generator. The comparison must allow both possibilities.

Our existence excludes histories that contain no relevant observers from an explanation of our observations. It does not by itself establish that the remaining weight concentrates on development from an initially unorganized state. Rival histories might contain compressibly generated observers at their boundary; many longer descriptions might collectively outweigh one shorter description. The selection claim therefore needs complete candidate histories and a rule for weighting their observers.

Within the class that begins below the required organization and later contains observers, development is necessary. Barren histories do not refute that implication: they fail its observer premise. What remains is to show why that developmental class receives the favored observational weight.

Does reversal require additional information?

A second route could favor forward experience even if reverse experience is possible: the reverse account might require more information or more processing to explain the same observations. These are separate claims. A longer computation can have a short description, while a long literal description can be quick to read.

Every unavoidable extra bit matters. If the complete reverse description needs k more bits than the forward description in the same code, its assigned weight is 2^(−k) times as large. One extra bit halves it; five reduce it by a factor of 32; ten by 1,024. This is an exact comparison of those descriptions under the stated measure.

The returning light, sound and heat make the proposed cost concrete: what specifies their coordination? Listing each incoming detail could be expensive. But a short rule might instead say to generate the forward history and reverse it, specifying those correlations together. For a finite history this supplies a compact description, although it does not establish physically initiated backward propagation. The question is whether any reverse account can match the forward account's economy, not how long one deliberately cumbersome account becomes.

But an added step in one construction does not prove a necessary extra cost in every construction. For a finite encoded history h and a fixed computable reversal R, a program for R(h) can run a program for h and apply R. Writing K for shortest program length, this gives K(R(h)) K(h)+c_R. If R reverses its own operation, the same argument in the other direction gives |K(R(h))−K(h)| c_R. This bounds the difference; it does not establish its sign or a positive minimum penalty for reversal.

The limitation is not merely linguistic. A rule that flips a bit is its own inverse: the reverse uses the same operation. More complicated reversible dynamics can likewise contain the inverse relation within a compact specification. A separate reversal command in our explanation may be absorbed by another program. The demonstration therefore needs a lower bound on unavoidable information, rather than an upper bound supplied by one extra instruction.

The deeper generative claim would be that a backward account must obtain information or organized structure that a forward account supplies directly, and cannot avoid that dependence while explaining the same evidence. In particular, if every realization of reverse experience first requires the forward development of its supporting structure, that would establish a priority of forward generation. It must be shown across the allowed realizations, rather than inferred from one implementation that deliberately builds the forward history first.

A useful comparison fixes the available starting information, the observations to be explained and the permitted resources. Any additional information needed only by one account must then be counted. A robust positive difference would

strengthen the explanation; a forward process merely stipulated as the reference language would not establish it. The evolutionary argument in section 4 asks a different question: which conditions support reliable learners at all. Neither a narrow set of successful conditions nor a longer calculation automatically establishes a longer shortest description.

6. Why entropy enters biological development

For observers like us, developing and maintaining organization uses available energy: energy differences that can do work. Cells take in food or light, build and repair structures, and release heat and waste. Local organization can increase while the entropy of the organism and its surroundings increases overall. This is why the growth of biological organization is compatible with the second law.

The relevant starting resource is a departure from equilibrium. In familiar thermodynamic models, equilibrium supplies no sustained gradient from which to extract work. A developmental account of life must therefore explain both the organizing dynamics and the usable differences they consume. Entropy production accompanies these actual processes; entropy is not another name for the organization they build.

The logical claim about development has wider scope than that physical account. An abstract rule can accumulate organized records reversibly, as the earlier memory example shows. To conclude that every possible mathematical observer must produce thermodynamic entropy would require an additional premise connecting the observer to a thermodynamic implementation. The present argument establishes necessary development from an initially insufficient state, and identifies the energy and entropy account needed for observers of our kind.

A simple description is not by itself a low-entropy starting state. For example, the string 010101… has a short generating rule. In a model that groups N-bit strings only by their number of ones, it lies in the largest macrostate when N is even. That macrostate has maximal counting entropy under the stated grouping. The short description and the large multiplicity coexist.

Thus the proposal must explain an available energy gradient, not infer one from the word simple. It can then connect the necessary growth of observer organization to the thermodynamic process that supplies it. This preserves the developmental argument while identifying the physical condition on which its entropy claim depends.

7. Comparing the directions

Suppose we begin with neither direction favored: a hypothetical coin flip. This illustrates an initial balance, rather than a measured fifty-fifty probability. The arguments presented here seek to shift that balance toward forward time. Even if reversal is possible, it may still receive less probability weight. Showing that a

reverse history exists answers an impossibility claim; it does not establish that observers should encounter it as often.

The proposed preference concerns a measure: a rule assigning probability weight to histories and to the observer experiences they contain. Extra information required to specify reverse histories could lower their weight under the simplicity measure. Restrictions on the conditions supporting reliable learners could lower the weight of reverse histories containing the observers whose evidence is being explained. The first route concerns description length; the second concerns how much of the allowed weight supports the relevant observers. Neither needs reversal to be impossible.

For a precise comparison, let W_F and W_R be the total weights assigned to forward- and reverse-oriented observer experiences compatible with the evidence being explained. If these alternatives exhaust the comparison and their weights have a finite, nonzero sum, their conditional probabilities are W_F/(W_F+W_R) and W_R/(W_F+W_R). The proposed preference is W_F > W_R. With equal starting weights, a lower weighted prevalence of the relevant observer experiences in the reverse class would establish it. This states what the arguments must explain; it does not assign a numerical advantage in advance.

The positive case combines rooted information production, the development of initially absent observer organization, evidence available before consequential action, and the energy gradients supporting biological growth. These considerations can support the same preference, but share assumptions and must not be multiplied as independent probability gains. A narrower successful set lowers its weight only under a specified measure; extensive coordination can sometimes have a short description. Thermodynamic entropy likewise does not automatically equal algorithmic description length.

Within the account developed here, the proposed weighting arguments favor forward development; no positive weighting argument for the reverse is supplied. A complete reversal of observer and environment remains the comparison to address. If reversal pairs the relevant observer experiences and preserves their weights, the totals are equal. Establishing the proposed preference therefore requires identifying which independently justified starting conditions, description costs or observer requirements break that equality. Merely reversing the coordinate label changes none of these relations.

What the production argument establishes

The missing final input excludes backward production under the stated rules. None of its steps becomes available merely because infinitely many other unavailable steps are specified. Forward generation, supplied with its first state, has no corresponding obstacle.

The endpoint argument asks whether the chain can obtain the information

needed to begin. The evolutionary argument asks which histories support reliable learners. These are distinct routes: impossibility under a production requirement, and lower probability weight for reversal among histories that remain possible. The probability route remains relevant even if the impossibility premise is not established.

The physical and experiential conclusion depends on two connections. First, experience must require the rooted information-production history just described. Second, the relevant history must have the first-but-no-last ordering. The simplicity and observer arguments seek to explain the appropriate initial conditions and subsequent development, but they do not remove either connection by definition.

A proof that reverse-oriented experience is impossible would have to exclude an observer instantiated in a jointly specified block or supplied with a state at an internal location. Calling such a structure mathematical does not settle whether it is conscious. Explaining why it cannot support the requisite succession is the work the generative account must do.

8. Conclusion

Forward propagation can begin from a supplied first state. Complete backward propagation through an endless future has no final state to supply its first input. If every step must receive already-produced information, the backward chain cannot begin. Applying that result to experience requires the paper's premise that experience needs such initiated production.

Possibility and probability are separate questions. Even if reversal is possible, every unavoidable extra specification bit halves its description weight under the stated measure. A reversed billiard break illustrates the coordination to be explained: scattered light, sound and heat must return as the balls reassemble. A short rule could specify that coordination, so its extra cost must be demonstrated rather than inferred from how strange the event looks.

Observer selection supplies a further necessity: organization absent at the beginning must develop if the history later contains minds requiring it. Biological development draws on energy gradients while total entropy increases. Reliable learning also requires evidence before action; an explanation arriving afterward cannot guide an earlier choice. A directional disadvantage must survive a comparison that allows the observer and its records to reverse too.

These arguments form a cumulative case for forward experience. The no-input result is exact under its production premise; the probability preference depends on the complete descriptions, their observer histories and their assigned weights. The hypothetical coin flip frames that comparison without pretending the size of the proposed advantage has already been calculated.

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